Wind Power Forecasting for Renewable Energy Systems Using LSTM Networks

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***Abstract*—Prediction of renewable energy production is a crucial aspect of ensuring stability in today's power systems. This paper proposes a short-term prediction model for the wind energy generation process in Germany based on an LSTM (Long Short-Term Memory) artificial neural network. Germany was chosen because of the high level of wind energy penetration and the existence of electricity datasets with a high degree of detail. Hourly wind energy generation datasets are used for training, and regression-based evaluation metrics are adopted. Experiment results show that the LSTM model performs well in predicting future values with low errors.**

I. Introduction

Increased integration of renewable sources like wind energy presents several operational challenges in contemporary power systems. Different from traditional sources, wind energy is quite unpredictable and varies depending on changing environment conditions. Therefore, precise forecasting of wind power generation becomes crucial for achieving sustainable operation, effective management, and minimizing costs.

As for the wind energy generation, it is characterized by non-linear dependencies and shows complicated temporal patterns like hour-to-hour variability, daily and seasonal dynamics. For proper modeling of such relationships, application of sophisticated methods has become inevitable. Among others, deep learning approaches have attracted much attention in forecasting wind power generation.

Specifically, in this study an LSTM network, a subtype of RNN, will be applied to forecast wind energy generation in Germany at hourly intervals. LSTM networks are capable of capturing long-term dependencies because of using memory cells and gates. Thus, they prove to be useful when dealing with time series problems.

Germany is chosen as the geographical region for this case study because of the high share of wind power generated there and access to relevant datasets. The objective of this study is to develop an accurate forecasting model and evaluate its performance using both quantitative metrics and visual analysis, providing insights into its effectiveness for real-world energy applications.

II. Dataset and Preprocessing

A. Dataset

The data that will be analyzed comes from a freely available dataset called Open Power System Data [1]. This type of data comes from actual power generation records from various European countries and has been successfully used in many time-series applications.

For this work, the variable `DE_wind_generation_actual`is chosen, which indicates the wind power production capacity in megawatts (MW) per hour in Germany. This data set consists of around 50,000 records gathered over several years, enabling the model to understand short-term and seasonal trends in wind power generation.

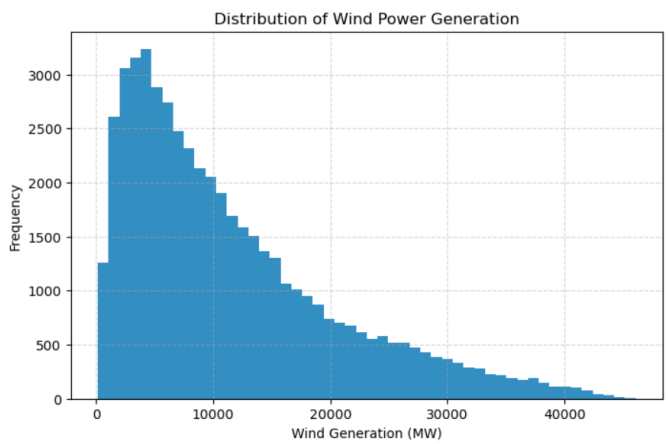

Fig. 1. Distribution of wind power generation with density curve. The skewed shape indicates high variability and non-linear patterns in the data.

The distribution along with the density curve is shown in Fig. 1. The distribution is positively skewed because of many low figures and very few highly skewed figures

showing high variance. The wind generation distribution is positively skewed, as the values occur mostly less than 20,000 MW but sometimes reach as much as 46,000 MW. Mean of the values is about 11,552 MW, which means that average values are moderately generated with high variance.

B. Preprocessing

The data set was pre-processed by deleting missing data and performing normalization on the data with the help of MinMaxScaler. The sliding window method was employed where the last 24 hours' data is used to predict the next hour.

TABLE I
SAMPLE DATA BEFORE AND AFTER PREPROCESSING

Timestamp	Before	After
2014-12-31 23:00	NaN	–
2015-01-01 00:00	AT: 69, BE: 9484	8852
2015-01-01 01:00	AT: 64, BE: 9152	9054
2015-01-01 02:00	AT: 65, BE: 8799	9070
2015-01-01 03:00	AT: 64, BE: 8567	9163

The data sample preprocessed and unprocessed are compared in Table I. To begin with, the dataset comprises a number of features from various countries with some missing values. As a result of preprocessing, there remains one feature, which is that of German wind power generation. The following preprocessing steps were applied:

- Removal of missing values
- Normalization using MinMaxScaler
- Sequence creation using a 24-hour sliding window

III. MODEL ARCHITECTURE

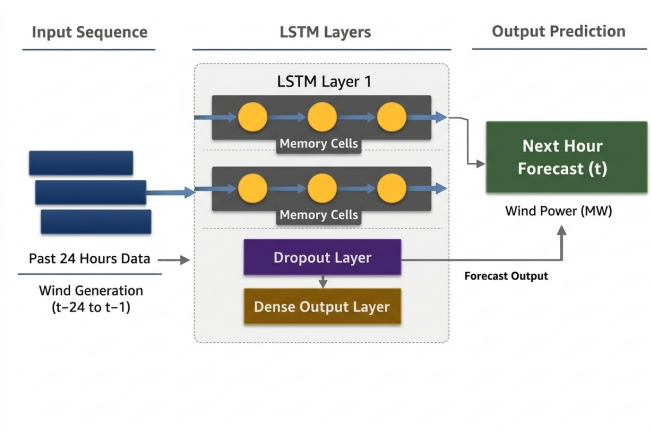

Fig. 2. LSTM architecture used for wind power forecasting.

The model consists of:

- LSTM layer with 50 units
- Dropout layer (0.2)
- Dense output layer

As per the proposed design, the architecture includes one LSTM layer having 50 nodes, and the subsequent layers include a dropout layer that has the rate 0.2 and a dense layer output layer respectively. LSTM layer is responsible for learning relations and patterns available in the input data sequence of past 24 hours. The purpose of dropout layer is to reduce overfitting in neural network. Finally, dense layer produces the predicted value of wind generation.

IV. TRAINING

The model was trained using:

- Optimizer: Adam
- Learning rate: 0.001
- Loss: Mean Squared Error (MSE)
- Epochs: 20
- Batch size: 32

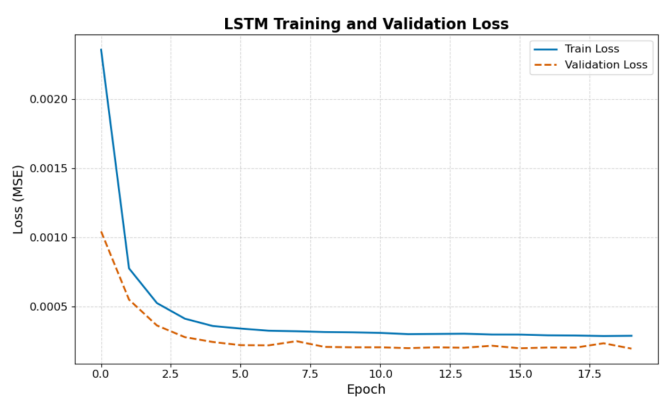

Fig. 3. Training and validation loss curves. The close alignment indicates no overfitting.

From Fig. 3, it can be seen that there is consistent reduction in training and validation loss, both converging to around 2.0×10^{-4}. From this, we can infer that the learning process of the model is effective. Also, since the curves for training and validation follow a similar trend, it implies that the model does not suffer from overfitting.

V. RESULTS AND EVALUATION

A. Performance Metrics

The model achieved the following results:

TABLE II
PERFORMANCE EVALUATION METRICS

Metric	Value
Mean Absolute Error (MAE)	487.75 MW
Root Mean Squared Error (RMSE)	693.72 MW
R^2 Score	0.9956

Since the upper limit for wind power generation is 46,000 MW, these errors are very insignificant, thus proving excellent prediction capabilities.

B. Prediction Analysis

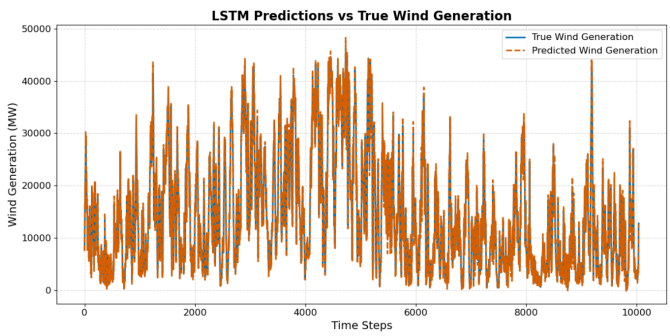

Fig. 4. Comparison of true and predicted wind generation values.

Figure 4 demonstrates how the predictions made by the model closely match the actual values, thus making accurate predictions on an overall trend pattern. The prediction pattern of the values follows the actual wind generation values to a good extent, including short-run fluctuations. However, minor variations are noted around the peaks.

C. Residual Analysis

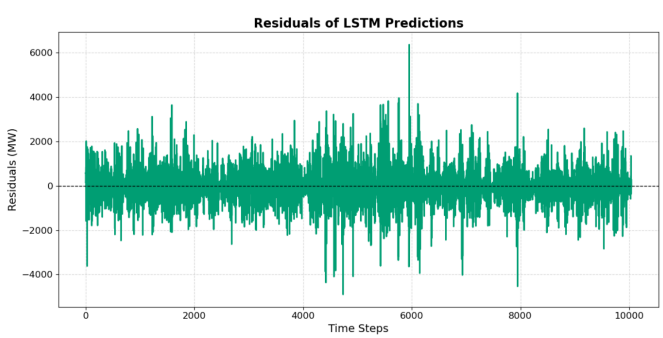

Fig. 5. Residuals over time for the LSTM model. The residuals are centered around zero with no clear pattern, indicating that the model does not exhibit systematic bias.

The residuals are zero-centered, and this shows that there is no systematic bias in the predictive capability of the model. The errors seem to be random through time with no discernible trend or pattern, meaning that the model generalizes very well to the test dataset. But some instances of large errors have been noted when there is high generation from wind power. This implies that it is hard to predict extreme values.

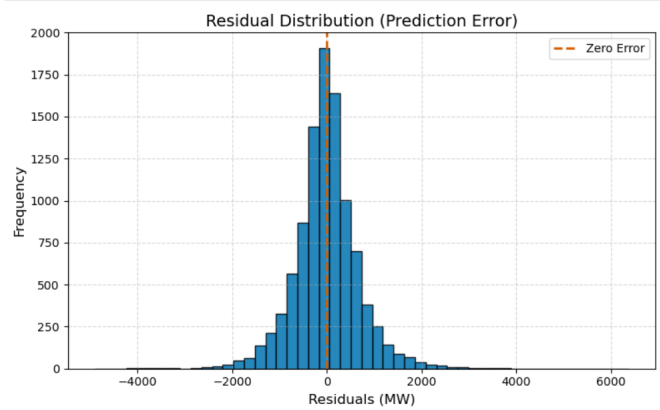

Fig. 6. Distribution of residuals. Most errors are concentrated around zero, suggesting that the model predictions are generally accurate with small deviations.

Figure 6 provides insight into how the errors of predictions vary. The majority of residuals tend to be centered around 0, which means that the forecasts of the model are accurate. As one can see from Table II, the mean absolute error of prediction is relatively low and equals 487.75 MW. It is important to note that this error is very small in comparison with the actual amount of generated wind, which can reach up to 46,000 MW.

It seems like the residuals tend to have a normal distribution, which means that the model predictions do not have any particular bias. In other words, the errors are not systematically overestimated or underestimated. Still, it is possible to state that some errors are bigger because of the increased scatter at the tails of the distribution curve, which corresponds to the fact that RMSE (693.72 MW) is higher than MAE.

In general, this phenomenon can be explained by the fact that bigger amounts of energy are usually associated with bigger errors. Thus, it should be concluded that the model makes quite accurate predictions in most cases but can sometimes face problems with forecasting large amounts of energy.

D. Prediction Accuracy

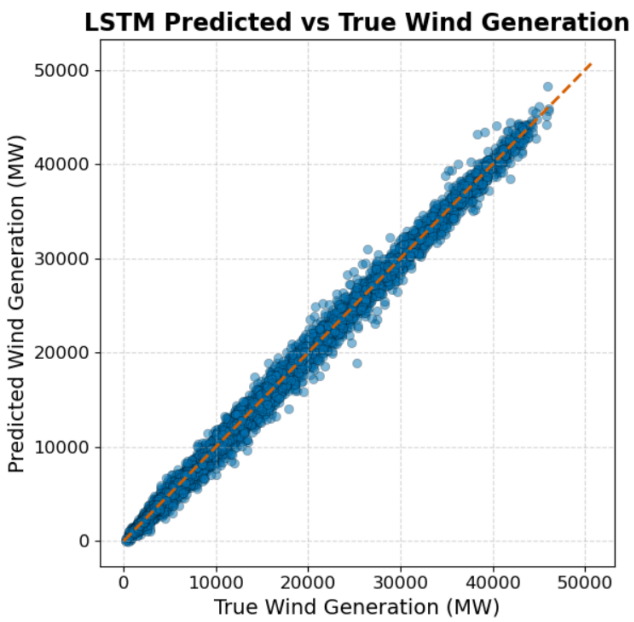

Fig. 7. Predicted vs true values. Points close to the diagonal indicate high accuracy.

As depicted in Fig. 7, most data points cluster around the diagonal line, suggesting high consistency between predictions and actual values. The dispersion is higher for large values, proving that prediction errors are more substantial when production is at its maximum.

E. Feature Distribution Analysis

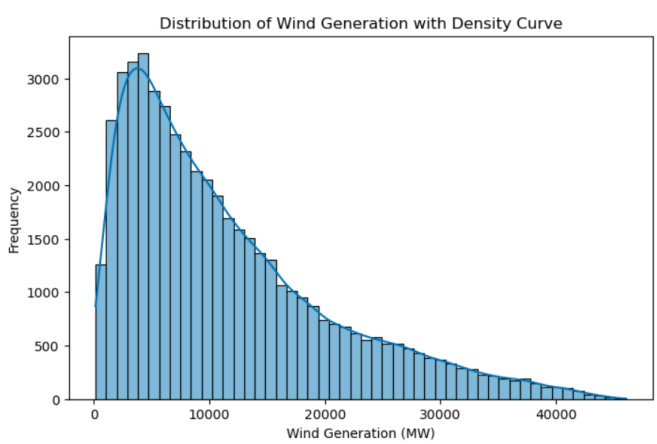

Fig. 8. Distribution of wind power generation with kernel density estimation (KDE).

The distribution of wind energy production along with its KDE is shown in Fig. 8. As evident from the figure, there is an asymmetric positive distribution of the values such that there is a considerable number of observations with low energy production up to 20,000 MW, whereas a smaller number of observations exist with high energy production up to 46,000 MW. There is a noticeable variability in the data, making LSTM neural networks suitable for the task.

F. Rolling Mean Comparison

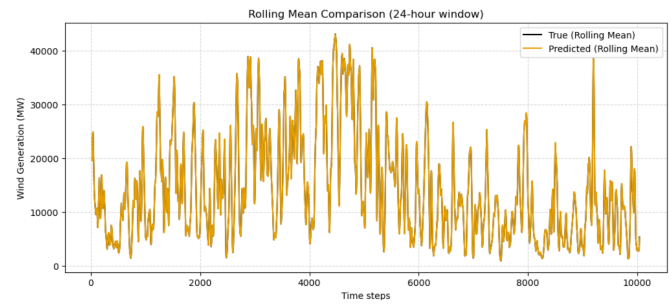

Fig. 9. Comparison of 24-hour rolling mean for true and predicted wind generation.

Figure 9 illustrates the comparison between the true value and the predicted value based on the 24-hour rolling average. The rolling average averages out short-term variations and emphasizes the inherent trend of the data. As can be seen from the figure, the predicted values align quite well with the trend of the actual values, implying that the model is indeed capable of predicting the temporal trends for wind power generation.

G. Performance Evaluation Metrics

Three regression metrics were applied quantitatively to measure the performance of the proposed model, such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2).

The model showed a Mean Absolute Error (MAE) of 487.75 MW, meaning that the error between actual data and predictions is quite low relative to the whole scope of generated wind capacity values, which can go up to about 46,000 MW. The RMSE value is equal to 693.72 MW, which is a bit bigger than MAE because of some outliers observed at peaks of wind generation.

The model produced R^2 score of 0.9956, which means that over 99% of wind power generation variance can be explained by the model.

All these results show that the performance of the model is high, with almost perfect results achieved.

VI. Discussion

The LSTM model successfully handles temporal relationships in the wind generation dataset. This is based on

the low MAE that shows prediction accuracy for most timesteps while the high RMSE shows some instances of error in prediction especially during extreme winds.

The analysis of the distribution shows that the dataset is both skewed and highly variable making it harder to predict. The model has performed very well despite this fact.

VII. Conclusion

In this case, a Long Short-Term Memory (LSTM) neural network is implemented for short-term prediction of the production of wind energy in Germany based on hourly data. The LSTM successfully models dependencies and general patterns in the input dataset. For instance, MAE and RMSE are estimated to be 487.75 MW and 693.72 MW correspondingly. These errors appear to be relatively small in relation to the maximal production that equals about 46,000 MW. As one can see from figures below, forecasted values approximate actual values well enough, while residuals fluctuate around zero, which means unbiased predictions. At the same time, the largest errors are observed when there is a sharp increase in the production of energy due to winds, which may become a problem when building such predictive models.

References

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